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Roll No

AU/ME-222-CBCS

B.E. III Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Theory of Machines and Mechanisms

Time : Three Hours

Maximum Marks : 60

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What is the Coriolis acceleration component? State and prove the 'Kennedy's Theorem' of three instantaneous centres.
b) Explain the procedure to construct Klein's construction to determine the velocity and acceleration of a Slider-Crank mechanism.
2. State and explain Three center in line theorem.
3. Explain how are Whitworth quick return mechanism and crank and slotted lever mechanism are different from each other?
4. Draw the profile of a cam to satisfy the following motion:
 - i) Follower to move outwards through 3cm with SHM during 160 degree of cam rotation
 - ii) Dwell for 20 degree
 - iii) Return with SHM in 160 degree and
 - iv) Dwell for 20 degree

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5. Define the term kinematic link and kinematic pair. Describe various inversions of a Slider-Crank mechanism giving examples.
6. A pair of gears, having 40 and 20 teeth respectively, are rotating in mesh, the speed of the smaller being 2000 r.p.m. Determine the velocity of sliding between the gear teeth faces at the point of engagement, at the pitch point and at the point of disengagement if the smaller gear is the driver. Assume that the gear teeth are 20° involute form, addendum length is 5 mm and the module is 5 mm. Also find the angle through which the pinion turns while any pairs of teeth are in contact.
7. In a four bar mechanism ABCD, points A and C are fixed points 30cm apart and AB, CD are bars 60cm and 70cm long respectively, which are connected by a rod BD which is 50cm long. If AB rotates with uniform speed of 60rpm. Determine Velocity of D when AB is perpendicular to AC and also when it makes 10° on either side of the perpendicular.
8. Define and explain terms:
 - i) Cam profile
 - ii) Base circle
 - iii) Pitch curve
